Math 133: Quiz 5

Friday, October 16, 2026 25 minutes 10 points

Name: _____

Show your work. Give exact answers unless an estimate is requested, and include units when interpreting rates or costs. A simple scientific calculator is allowed, but is not needed.

1. (3 points) Find each derivative. You may use differentiation rules.

   (a) $g(x) = 4x^3 - 5x + 7$.

   (b) $r(x) = \dfrac{6}{x^2}$, $x \neq 0$.

   (c) $s(x) = 3\sqrt{x} + 2x$, $x > 0$.

2. (2 points) Find an equation of the tangent line to the graph of $f(x) = x^2 - 3x + 2$ at the point $(1, 0)$.

3. (2 points) A company's total cost of producing q notebooks is $C(q)$ dollars. The function C is differentiable, and

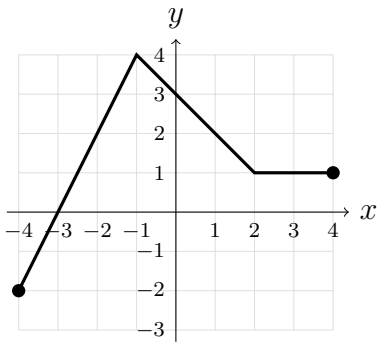
$$C(100) = 2500, \qquad C'(100) = 12.$$

(a) Interpret $C'(100) = 12$ in this situation. Include units.

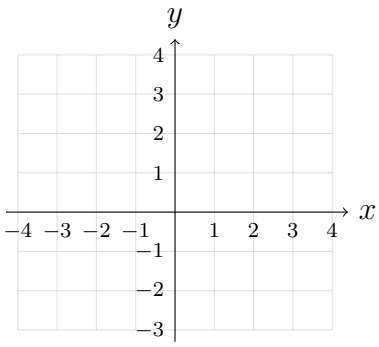
(b) Write the tangent-line approximation $L(q)$ for total cost near $q = 100$, and use it to estimate $C(103)$.

4. (3 points) The graph of a piecewise linear function h is shown on the left. Sketch h' on the right for $-4 < x < 4$. Use open circles where appropriate, and state the x-values in this interval where h' does not exist.

$y = h(x)$



$y = h'(x)$



h' does not exist at: _____